Mohammad Savari

PhD Candidate — Network Science — System Dynamics
University of Ottawa

Professional Summary

- **PhD Candidate** in Mechanical Engineering (2021–2026) at University of Ottawa
- Researching **Social Contagion** and **Complex Networks** under Prof. Roland Bouffanais
- Focus: Understanding information propagation and collective behavior in interconnected systems
- **Full admission scholarship** and **merit scholarship** recipient
- Experienced in **network science**, **systems dynamics**, and **computational modeling**

"Exploring how ideas and behaviors cascade through complex networks"

Current Positions

- **Project Manager** (Sep 2024–Present)
  - Centre for Entrepreneurship and Engineering Design, Ottawa
  - Managing 4 student groups developing accessibility devices
  - Coordinating procurement and monitoring design timelines
- **Search Engine Evaluator** (Jun 2024–Present)
  - TELUS International
  - Analyzing search relevance, accuracy, and content quality
  - Identifying trends to improve algorithms
- **System Engineering Intern** (Sep 2023–Mar 2024)
  - Airshare Inc., Ottawa
  - Developed system models and enhanced communication frameworks

Academic Background

- **Complex Networks Researcher** (May 2025–Present)
  - Telfer School of Management, Ottawa
  - Web scraping, NLP analysis, network visualization using Python and GEPHI
  - Identify and characterize sub-networks, communities, and topological properties in complex global networks using network science methodologies
- **Social Contagion Researcher** (Sep 2021–Present)
  - Faculty of Engineering, University of Ottawa
  - Developed new methods for integrating system dynamics in social contagion analysis
  - Created heterogeneous free-scale networks with tuneable clustering
  - Performed distribution likelihood analyses
- **Teaching Assistant** (Jan 2022–Present)
  - Courses: Mechatronics, Systems Dynamics, Biomedical Systems Dynamics, Introduction to Product Development, Machine Learning in Control, Robotics

Education

PhD in Mechanical Engineering (2021–2026 Expected)

- University of Ottawa, Canada
- **Thesis:** "Fostering Consensus using Social Contagion"
- Direct-PhD path from bachelor's degree
- Full admission scholarship and merit scholarship

Bachelor of Science in Electrical Engineering (2014–2018)

- Foundation in control systems, electronics, systems design, and signal processing

Technical Skills

Programming & Analysis

- Python
- R
- MATLAB
- LabVIEW
- Statistical analysis

Embedded Systems

- Atmel chips
- Arduino
- PLC platforms

Research & Technical

- Complex systems modeling
- Social contagion analysis
- Network development
- Data collection, processing & analysis

Other Skills: Project management, Leadership, Problem-solving, Communication, Collaboration

Research Focus

PhD Research Theme: Fostering Consensus using Social Contagion

Key Research Areas:

- How social contagions shape collective consensus on scale-free networks
- Systems dynamics in information propagation
- Heterogeneous network modeling and analysis
- Mechanisms of network formation and evolution

Publications:

- **Submitted (2025):** "How social contagions shape collective consensus in the presence of scale-free networks" – Savari, M., et al.
- **Conference (Jul 2025):** "Leveraging Social Contagion to Foster Consensus in Collective Decision-Making" – ICCS, Singapore (abstract paper)

Key Achievements

- **Direct PhD Entry** from Bachelor's degree – Recognition of academic excellence
- **Full Admission & Merit Scholarships** from University of Ottawa
- **Novel Research Methodology** for integrating system dynamics in social contagion analysis
- **Multi-disciplinary Expertise** bridging network science, systems dynamics, and computational modeling
- **Diverse Leadership Experience** in project management, research coordination, and community service
- **International Conference Presentation** at ICCS 2025, Singapore

Get in Touch

Mohammad Savari

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Interests:

- Systems Dynamics & Collective Behavior
- Computational Modeling & Data Science
- Complex Networks & Network Science
- Social Contagion & Information Diffusion

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MohammadSavari.github.io

Thank You

Exploring the Dynamics of Complex Networks

"In complexity lies the key to understanding collective behavior"